RESEARCH PROPOSAL

Title: ***In-Vermis* Ultra-Long DNA Synthesis**

Project coordinator: **Prof. Oded Rechavi**

1. Abstract

The synthesis of long DNA sequences is a crucial yet challenging task in synthetic biology and related fields, hindered by the high costs and labor intensity of current technologies. This project proposes a novel approach, leveraging the natural formation of DNA arrays in *Caenorhabditis elegans* (*C. elegans*) nematodes to synthesize ultra-long DNA sequences efficiently. *C. elegans*, a well-established model organism, can concatenate injected DNA into extrachromosomal arrays (ECAs) millions of bases long. We aim to exploit this mechanism to create a one-step, cost-effective "cloning machine" for synthesizing and amplifying error-free ultra-long DNA sequences.

Our research objectives include: **1)** Deciphering the rules governing ECA formation by injecting varying combinations of DNA sequences into *C. elegans* gonads and employing advanced sequencing and computational analysis on resulting ECAs; **2)** Understanding and leveraging the role of specific DNA repair mechanisms in ECA formation, using mutants in repair pathways; **3)** Establishing an integrated system for long DNA synthesis, including software for designing the requisite short DNA sequences and an automated system for injecting them. **The significance of this work lies in its potential to overcome existing barriers in DNA synthesis. Our project represents a novel intersection of genomics, synthetic bioengineering, and model organism research, with the ability to significantly impact our utilization of biological systems for engineering purposes.**

2. Scientific background and state of the art

DNA synthesis is fundamental for many sectors of research and commercial activities, such as synthetic biology, therapy, data storage, and nanotechnology. However, synthesizing long DNA sequences is currently a major bottleneck, as contemporary technologies for DNA synthesis, such as molecular assembly and cloning methods, template-independent enzymatic synthesis, microarray, and rolling circle amplification techniques are still unaffordable at large scale, are extremely laborious, and require numerous steps, iterations, and quality control (1). The lack of efficient long DNA synthesis methods has stalled the progress of these fields and remains a main obstacle for multiple advancements in synthetic biology, from research to drug development, DNA-based computing, and DNA storage.

C. elegans is a small, free-living roundworm that has become one of the most studied and important model organisms in scientific research. For over three decades, it has been well-established that when DNA is injected into the worm's gonads (plasmids, for example), the worm concatemerizes the injected DNA and forms gigantic, repetitive, extrachromosomal "arrays" (ECAs), which are typically 1-3 million base pairs long (2). Previous studies utilized traditional techniques such as gel electrophoresis, southern blots, and "DNA paint" to study this phenomenon and probe the ECAs composition. However, the exact mechanisms shaping and dictating the formations of ECAs and the exact DNA sequence created have yet to be fully elucidated. It has been conceptually challenging to determine the exact DNA sequence of ECAs because of their length and highly repetitive nature, preventing the use of standard short-read sequencing techniques such as Illumina.

Emerging third generation long-read sequencing technologies enable sequencing of single molecules of DNA without amplification or fragmentation. These methods can be used to overcome this limitation and gain new insights on the exact sequences of the ECAs. The advancement of computational tools opens new possibilities to analyze these novel datasets and gain in depth understanding of the rules that guide the formation of ECAs in ***C. elegans*** following microinjection of different combinations of short DNA constructs.

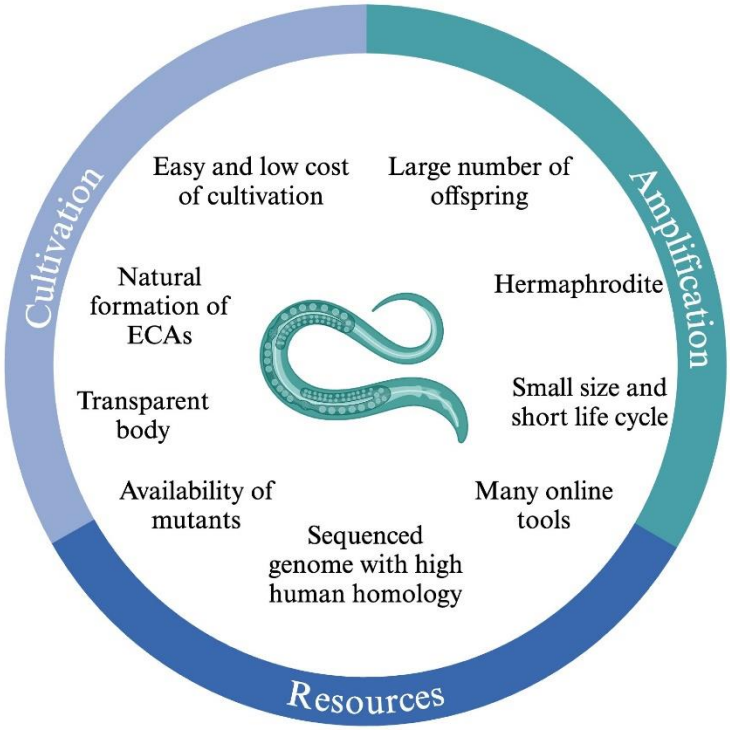

Figure 1*: Advantages of* ***C. elegans*** *for ultra-long DNA synthesis*

In this project we suggest combining these state-of-the-art technologies to turn the worm into a potentially revolutionary *in-vermis* "factory" (*in vermis* means "in worm") for accurate and cost-effective synthesis of ultra-long DNA sequences (1-2 Mb long). The worms also accurately amplify ECAs across generations, and the worms are hermaphroditic, so a single worm carrying the correct ECA can give rise to hundreds of progenies carrying the same exact ECA (**Figure 1**). **Our goal is to develop a one-step procedure, the ideal "cloning machine", where one simply introduces thousands of short DNA sequences into the worm, which will then use its endogenous DNA array formation process, to synthesize error-free and huge DNA sequences, and to amplify the resulting DNA using its natural replication machinery.**

3. Research objectives and specific aims

Objective 1 – Decipher the rules of ECA formation in *C. elegans*. We will inject an increasing number (2-100) of different 5-10 Kb DNA sequences into wild type *C. elegans* gonads. We will first use split-reporter experiments to identify successful predetermined DNA recombination (only upon successful recombination the reporter will fluoresce) and then use our advanced long-read DNA sequencing and mapping techniques and bespoke computational tools to analyze the resulting ECAs. We will test the effects of different homologies, ranging from 0 to over 800 bases, on the recombination and ECA formation. This objective will elucidate the roles of different sequences and of homology-directed repair in ECA formation and determine the ideal degree of homology between injected segments for subsequent experiments.

Objective 2 – Understand the role of DNA repair mechanisms in ECA formation. Using nematodes with loss-of-function mutations in different DNA repair genes, we will repeat the injection experiments elaborated in Objective 1. Comparing ECAs formed in wild type nematodes and in mutants with enhanced homology-directed repair and reduced non-homologous end joining will allow us to harness homology for desired ECA formation. Using our advanced computational tools, we will use the data collected in the two objectives to decipher the mechanisms enabling the design and production of >1 Mb DNA sequences by injecting short DNA sequences into *C. elegans*

gonads. This objective will culminate with our ability to synthesize ECAs from 10 homology-sharing sequences comprised of only the desired homologous recombination connections.

Objective 3 – Establish a one-stop-shop system for *in-vermis* long DNA synthesis. Based on the information and understanding acquired in objectives 1-2, we will develop a proof-of-concept system to synthesize long DNA sequences including an easy-to-use software to translate the requested long sequence to the needed short sequences to be injected into the worms. We will streamline the procedure using the previously described automated robotic system (3) for plasmid injection and incorporate hybridization capture and exonuclease treatment into the process to extract the synthesized long DNA sequences from the worms. This objective will culminate with our ability to use *C. elegans* to synthesize and isolate entire predetermined genomes of viruses, mitochondria, and small bacteria (Mycoplasma) inside the worm.

4. Detailed description of the proposed research

C. elegans assemble DNA arrays from injected short DNA sequences without the need for further induction, yet the exact rules of recombination/ligation are yet to be fully elucidated. Our preliminary results and previous research in the field indicate that the key to correct recombination is the homology between the sequences forming the desired ECA. Our preliminary results further suggest that non-specific joining of non-overlapping fragments can be minimized by neutralizing non-homologous DNA repair. Thus, to understand the rules governing ECA formation, we want to understand the exact contribution of different repair mechanisms involved in forming the array. To understand the rules of recombination and optimize accurate ECA creation, we will: 1) generate arrays of increasing complexity by injecting small numbers of fragments, with varying degrees of homology and different lengths. We aim to perform the equivalent of "Gibson assembly" (4) in the worm, where homologous sequences in the end of one DNA fragment and the start of the following DNA fragment lead to connections between fragments in a desired order. 2) Manipulate the worms' DNA repair pathways via loss-of-function mutations in DNA repair genes, and then create ECAs in these worms and see the effects on ECA formation. Ultimately, our goal is to inject hundreds of pieces of short DNA sequences with specific regions of homology that lead to the formation of a pre-determined ECA longer than 1Mb.

As part of this project, we will rely on three methods to examine the formation of ECAs in the experiments above. To sequence the de novo ECAs and examine the fragment-to-fragment recombination, we are using Oxford Nanopore PromethION Sequencing. To investigate the macro-structure of the ECAs and help us create an accurate de-novo assembly of the arrays, we complement the third-generation sequencing input with Saphyr optical genome mapping. To enable high throughput examination on multiple combinations, we will use split-reporter experiments, using multiple fragments of DNA, each of them not encoding anything, but when recombined in the correct manner they create a visible phenotype in the worms, such as fluorescence (we already obtained promising preliminary results with all 3 technologies. See below). This gives us a cheap and scalable method to screen accurate array formation in various conditions and genetic backgrounds. We can even scale it up to encode multiple, distinct phenotypes upon the accurate recombination of many fragments (e.g. multiple fluorescent proteins, or other visible phenotypes).

To analyze the data, we developed bioinformatic methods to reliably detect and classify desired vs undesired recombination of DNA fragments in the ECAs. As part of the development of the technology for high-throughput formation of predetermined long sequences of DNA in single microinjection of large number of short DNA

sequences, we plan to rely on previously demonstrated methodologies for robotic injection of plasmids into the ***C. elegans*** gonads.

5. Significance, innovation and potential benefits of the proposed research

Advances over the last few decades in sequencing and synthesis of DNA sequences have improved our ability to understand and engineer biological systems and led to the development of groundbreaking technologies for the design, assembly, and manipulation of DNA encoded genes, materials, circuits, and metabolic pathways. Synthetic biology is emerging as an important discipline with the potential to reshape many academic and industrial applications including the creation of novel therapeutics, materials, biosensing, and manufacturing capabilities. Despite advancement in the field, the technological gap still exists and most of the methodologies for synthesizing long DNA strands are similar, multistep processes using the same starting materials and relying on synthetic chemical and enzymatic reactions. This is a major bottleneck for many sectors. For example, the synthesis of the first eukaryotic genome, *Saccharomyces cerevisiae*, is an ongoing international effort involving numerous research groups that has resulted in the synthesis of the yeast's 16 chromosomes that range from 0.2-2Mb in length for a total of 12 Mb per genome. Each chromosome was synthesized by combining 100s of 'mini-chunks' of 2-4kb DNA that were combined sequentially into larger 'chunks' and again into 'megachunks' which were inserted into a YAC/BAC vector backbone, and in some cases meiotic recombination-mediated assembly was used to recombine all these pieces into one chromosome (5).

The reported cost of the synthetic yeast chromosomes is an average of $0.10 per base pair, translating to $1.2 million to synthesize one *S. cerevisiae* genome (6). This cost represents the cost per base when synthesizing segments < 3000 bp which are then combined using various techniques. The cost per base of synthesizing longer segments in one step increases dramatically with length and can reach more than ten times that of shorter DNA fragments, amounting to over $10 million for one yeast genome. Alongside the high cost per base is the high cost of labor due to the time required to synthesize these long segments of DNA by multi-step, technologically challenging procedures.

The possibility of harnessing and manipulating natural processes for the synthesis and amplification of novel, ultra-long DNA strands presents an opportunity for breaking the DNA production barrier and propelling synthetic biology to new levels of applicability. *C. elegans* has already played a significant role in the scientific discoveries of the past few decades as one of the leading model organisms to study many aspects of biology, and innovation in worm genomics were pivotal for whole genome sequencing (7). DNA injected into the *C. elegans* germline forms de-novo centromeres at high frequency, providing unique access to a process that occurs with extremely low frequency in other systems (8).

The worms not only assemble the DNA construct for us, they also accurately propagate and amplify it across generations, and worms carrying the ECA are easy and cheap to grow and upscale. The worms are self-fertilizing hermaphroditic, so a single worm carrying the correct ECA can give rise to hundreds of progenies carrying the same exact ECA (no need to set up crosses). Some worms produce identical arrays in all their cells, however, since extrachromosomal arrays are segregated by mitosis and inherited in a non-mendelian manner, we can select mosaic animals which contain varying numbers of arrays and degrees of gene expression within different cells. Thus, this unconventional inheritance holds the potential of producing diverse long DNA sequences and using ECAs as

reliable tools for gene expression. It is necessary to understand their mechanism of inheritance and more finely control the rate of ECA transfer and accumulation within cells for ECAs to meaningfully contribute to synthetic biology. As we are experts in unconventional modes of inheritance (9-11), we are uniquely positioned to study ECAs and begin harnessing their possibilities also in the biotechnological industry by turning the microscopic nematodes into natural mini-factories for high-throughput and accurate production of long DNA strands. In addition, establishing *C. elegans* as a platform for ultra-long DNA synthesis could serve as a proof-of-concept for the utilization, in the future, of yet to be fully understood mechanisms of ECA generation and manipulation of in other organisms such as *Plasmodium falciparum* (12)**,** *Paramecium* (13) and *Oxytricha trifallax* (14)**.**

6. Applicability

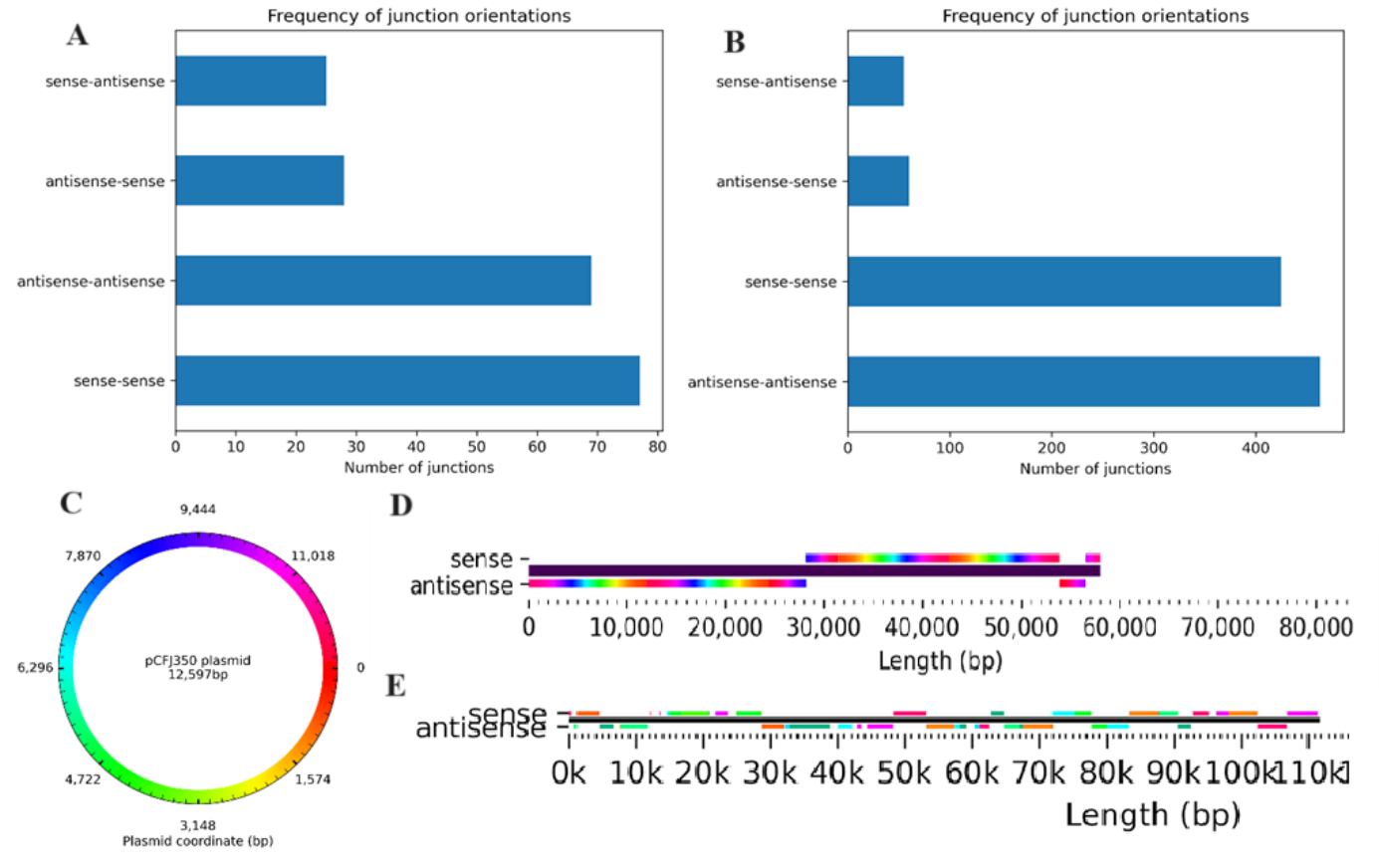

***Figure 2**. Plasmid connections in ECAs. **A)** There is a strong bias (~75% of connections) toward same-orientation connections (sense-sense, antisense-antisense) between a plasmid with sticky ends and **B)** a stronger bias (~90% of connections) in a plasmid with blunt ends. **C)** The sticky-end plasmid represented as a colormap to easily distinguish plasmid coordinates. **D)** Sense and antisense alignments to a ~60 kb sequence of the ECA. Same orientation connections tend to form between the end coordinate of one plasmid to the start coordinate of another, "reclosing" the plasmid. Opposite orientation connections tend to form at the same coordinate on both plasmids. **E)** Alignments of plasmids to a 110 kb section of an ECA that formed from 43 different plasmids. Each unique color represents a different plasmid.*

C. elegans is one of the most widely used model organisms in biology today. Since its establishment as a model organism, research conducted using the worm has significantly contributed to discoveries in fields such as developmental biology, genetics, neuroscience, and aging, and led to the award of six Nobel prizes. The popularity of the worm as a model organism can be attributed to several key features, including its ease and low cost of cultivation, short generation time, transparent body, and well-defined nervous system. *C. elegans* was the first animal to have its genome sequenced and has been extensively studied. These characteristics position the nematode

as ideally suited for utilization as a powerful tool also for biotechnology and synthetic biology when the right application is developed.

In our preliminary study, we have already demonstrated the applicability of using our methods to begin deciphering the mechanisms shaping the formation of ECAs. We successfully used long read sequencing technologies to sequence arrays of different compositions. We created arrays of increasing complexity, starting with arrays that are made of a repetitive single short DNA sequence, moved on towards arrays that are made of two different DNA sequences, and finally successfully sequenced and assembled an array which is made of **43** different DNA plasmids. We injected *C. elegans* worms with a plasmid either with "blunt ends" or "sticky ends" and characterized the way in which the plasmids connected to form an extrachromosomal array. We found that there is a bias toward same-orientation connections, which can be seen more strongly in the blunt ends scenario (**Figure. 2A** and **2B**). Also, junctions in both the same and opposite orientations (sense-sense vs. sense-antisense) tend to occur at the same coordinate in the two plasmids (**Figure. 2C and 2D**). We mapped the alignments of 43 different plasmids with varying degrees of homology to the ECA that they formed and visualized their connections (**Figure. 2E**). In preliminary studies of array formation in worms that are mutants in non-homologous end joining (NHEJ), we found that it was much harder to obtain a stable ECA, suggesting the importance of NHEJ repair in forming arrays and pointing toward expression of homology-directed repair pathways as a means to encourage homology-based connections. Furthermore, preliminary results with the split mCherry reporter described in Objective 1 yielded fluorescence in the 503 bp condition, pointing to homology-directed repair as an important mechanism used by the worm in array formation. In addition, we have recently published multiple software tools aimed at simplifying computational analysis for non-specialists, demonstrating our expertise in the development of this type of tool (15) (16). Our unique knowledge and experience in using the *C. elegans* for multiple and diverse types of research, from non-mendelian epigenetic inheritance to neuroscience research, position us as ideally suited to accomplish this high-risk, high-gain concept of establishing a novel one-stop-shop technology for in-vermis ultra-long DNA synthesis, that could have transformative effects on the entire field of synthetic biology.

7. Work plan and Gantt

Objective 1 – Decipher the rules of ECA formation in *C. elegans*.

Task 1.1 (Start: month 1, End: month 3): Identify optimal homology length for ECA formation.

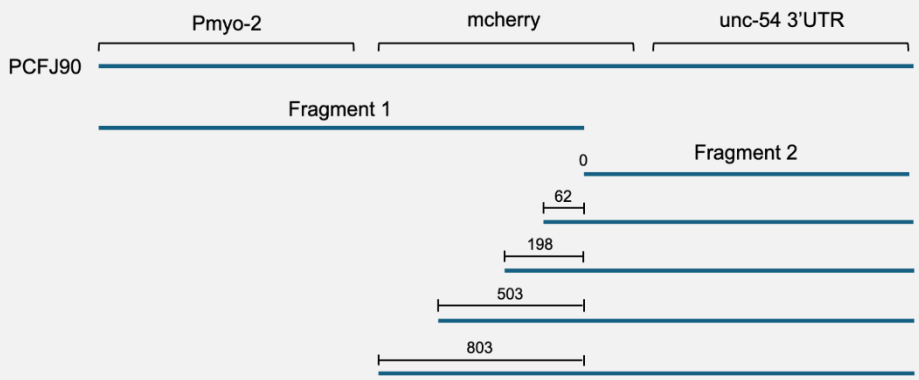

Figure 3. *Split mCherry reporters. The PCFJ90 plasmid split into five sets of two fragments. The sets range from 0-803 bp of overlap between the fragments. Only error-free recombination of the two fragments will lead to mCherry expression.*

We will use the PCFJ90 plasmid [Pmyo-2::mcherry::Punc-54 3'UTR] to create 5 sets of split mCherry reporters with varying degrees of overlap (**Figure 3**). The overlaps will range from 0 bp (no overlap) to 803 bp. The two halves of the split reporter will be injected into the worms so that only successful, error-free recombination

(homology-directed repair) will lead to fluorescence from the intact mCherry gene. We will count the number of fluorescent progenies in each overlap condition to determine whether we have successful recombination. This will allow us to determine a cutoff point of necessary homology length under which we see no meaningful successful recombination. Our preliminary results using 503 bp overlap yielded fluorescence, attesting to the validity of the method and the role of homologous recombination in array formation as the injected split reporter was correctly repaired.

Table 1. Fluorescence- informed determination of homology needed for successful recombination.

Overlap (bp)	**0**	**62**	**198**	**503**	**803**
Fluorescence	yes/no	yes/no	yes/no	yes/no	yes/no

Task 1.2 (Start: month 4. End: Month 6): Sequence the successful recombination products.

If the 0 and 62 bp overlap conditions result in fluorescence, we will use qPCR to quantify the number of successful recombination events in each condition and compare. Since the recombined product in all conditions is the same, we will use CRISPR with the same five guide sequences for all overlap conditions that yielded fluorescent progeny to fragment the recombined product in the resulting arrays into four segments of 200-250 bp each (**Figure 4**). On these fragments we will perform Illumina sequencing with unique molecular identifiers to determine the sequences of recombined fragments and their relative occurrence in each overlap condition.

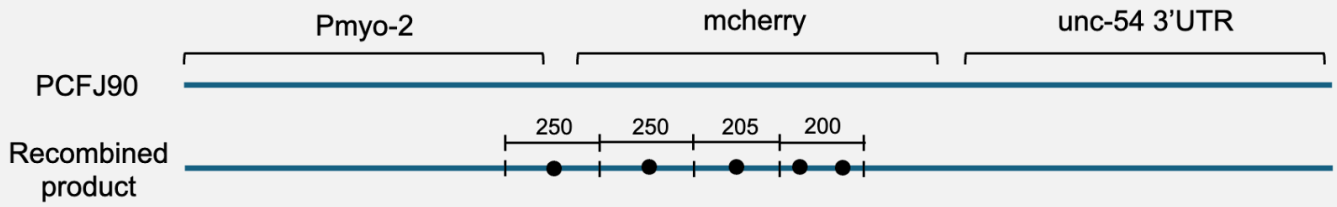

Figure 4. *CRISPR splicing of recombination region in each of the ECAs. Lines on the recombined product represent CRISPR cut sites and dots represent the ends of each of the segments in the five overlap conditions. Four segments of 200-250 bp will be sequenced with unique molecular identifiers using Illumina.*

Task 1.3 (Start: month 7. End: month 12): Increase the number of homologous subunits with the optimal overlap length.

Having discovered the optimal lengths of homology between two injected sequences, the previously described sequencing analysis will be performed on an ECA formed from three injected sequences sharing homologous regions. The resulting ECA will also be sequenced using third generation Oxford Nanopore Technologies and its sequence will be determined using genome assemblers. The connections that formed between the injected segments will be analyzed and further optimized to enhance the amount of successful recombination.

Objective 2 – Understand the role of DNA repair mechanisms in ECA formation.

Task 2.1 (Start: month 13 End: month 16): Study the effects of mutations in DNA repair mechanisms.

Optimized three-segment injection mixtures with the optimal lengths of homology identified in Objective 1 will be injected into DNA repair mutants defective in the non-homologous end-joining DNA repair pathway or overexpressing genes in the homology-directed repair pathway (see **Table 2**). The resulting ECAs in mutants will be compared to those in wild type nematodes and the effect of the mutation on array formation will be examined.

Table 2. Candidate DNA repair genes for mutant creation.

Human gene	Worm ortholog	Pathway
Lig4	*lig-4*	canonical NHEJ (non-homologous end-joining)
Ku70 / Xrcc6	*cku-70*	NHEJ
Ku80 / Xrcc5	*cku-80*	NHEJ
Nbs1	*nbs-1*	DNA end resection in alternative NHEJ and HR (Homology-directed Repair)
Fan1	*fan-1*	5'-3' nuclease implicated in interstrand cross-link repair and replication fork stability
Polq	*polq-1*	MMEJ (Microhomology-Mediated End Joining)
Parp1	*parp-1*	Promotes MMEJ
Msh2	*msh-2*	MMR (Microhomology-Mediated Repair)
Mlh1	*mlh-1*	MMR

Task 2.2 (Start: month 16. End: month 18) Identify desired recombinations and create multi-gene mutants for optimal array formation.

The resulting arrays in repair mutants will be sequenced using Nanopore. The mutants that result in the most desirable ECA connections will be crossed to create multi-gene mutants that lead to high occurrence of array formation using mostly homology-directed repair.

Task 2.3 (Start: month 19 End: month 30): Create arrays with only desired connections from homology-directed repair.

Once the most effective mutants are established and crossed, we will use advanced statistical techniques such as Bayesian Optimization to determine a minimal set of experiments required to maximize the accuracy and yield of ECA formation. These experiments will test different combinations of experimental parameters, such as number of fragments, fragment size, overlap length, GC content, total DNA quantity, and expression/activity levels of different DNA repair genes, to optimize the accuracy of homologous repair between the injected fragments, while maximizing the success rate of ECA formation.

Objective 3 – Establish a one-stop-shop system for in-vermis long DNA synthesis.

Task 3.1 (Start: month 31 End: month 33): Develop software for injection mix prediction.

For applicability and accessibility, we plan to translate our findings into an easy-to-use software to help predict injection mixtures. The input to the software will be the desired long DNA sequence that will be synthesized, and the output will be its building blocks in the form of 3-10 kb segments and each of their concentrations in the injection mixture. The software will take into account gene continuity, restriction enzyme locations, and existing

areas of homology within the desired sequence, as well as the optimal ranges of various factors determined during sub-aim 2.3. The software will also be able to predict the probability of success for a given experiment and suggest modifications to improve the chance of success.

Task 3.2 (Start: month 28. End: month 33): Usage of robotic injection system (outsource).

A robotic injection system has been previously described to streamline *C. elegans* transformation via microinjection (3). The system increases the speed and success of injection 23-fold and 1.6-fold, respectively, compared to an experienced human injector. We will hire an engineer to build a system based on this design so that injections can be easily and quickly performed to form predicted extrachromosomal arrays.

Task 3.3 (Start: month 31. End: month 33): Develop ability to capture array.

In order to use the *in-vermis* synthesized DNA, the arrays must be extracted and isolated from the worm's genomic DNA. To develop the DNA extraction technique, we will rely on restriction enzyme digestion, a newly developed tool that allows for the planning of injected sequences such that the *C. elegans* genomic DNA can be digested while leaving the array intact. Next, we will eliminate the digested sequences using short sequence elimination and the remaining can be further isolated using an agarose plug.

Task 3.4 (Start: month 34 End: month 36): Proof of concept of prediction software + array capture and sequencing.

As a proof of concept of the array formation pipeline, we will construct the genome of the 1.83 Mb *Haemophilus influenzae* bacterium from an injection mixture predicted using our prediction software. We will inject the array using the robotic injection system, capture it using restriction enzyme digestion and agarose plug extraction, and sequence it with Nanopore to ensure that the resulting genome was recombined as desired *in-vermis*.

Team:

The molecular biology research will be led by a postdoctoral and a PhD students who will jointly conduct the cloning and crosses of the ***C. elegans***. The postdoctoral student will also be responsible for developing the bioinformatics necessary for the project.

#	Task	6	12	18	24	30	36
1	1.1						
2	1.2						
3	1.3						
4	2.1						
5	2.2						
6	2.3						
7	3.1						
8	3.2						
9	3.3						
10	3.4						

8. Bibliography

1. *Hoose, A., Vellacott, R., Storch, M., Freemont, P. S., & Ryadnov, M. G. (2023). DNA synthesis technologies to close the gene writing gap. *Nature Reviews Chemistry*, *7*(3), 144-161.
2. * Stinchcomb, D. T., et al. (1985). Extrachromosomal DNA transformation of Caenorhabditis elegans. *Molecular and Cellular Biology, 5*(12), 3484–3496.

3. *Dong, X., Song, P., & Liu, X. (2021). Automated robotic microinjection of the nematode worm Caenorhabditis elegans. *IEEE Transactions on Automation Science and Engineering, 18*(2), 850-859.

4. *Gibson, D. G., et al. (2010). Creation of a bacterial cell controlled by a chemically synthesized genome. *Science, 329*, 52-56.

5. Zhang, W., et al. (2023). Manipulating the 3D organization of the largest synthetic yeast chromosome. *Molecular Cell, 83*(23), 4424-4437.

6. *Richardson, S. M., et al. (2017). Design of a synthetic yeast genome. *Science, 355*(6329), 1040-1044.

7. The C. elegans Sequencing Consortium. (1998). Genome sequence of the nematode C. elegans: A platform for investigating biology. *Science, 282*, 2012-2018.

8. Yuen, K. W., Nabeshima, K., Oegema, K., & Desai, A. (2011). Rapid de novo centromere formation occurs independently of heterochromatin protein 1 in C. elegans embryos. *Current Biology, 21*(21), 1800-1807.

9. Houri-Zeevi, L., et al. (2020). Three rules explain transgenerational small RNA inheritance in C. elegans. *Cell, 182*(5), 1186-1197.

10. Rieger, I., et al. (2023). Nucleus-independent transgenerational small RNA inheritance in Caenorhabditis elegans. *Science Advances, 9*, eadj8618.

11. Toker, I. A., et al. (2022). Transgenerational inheritance of sexual attractiveness via small RNAs enhances evolvability in C. elegans. *Developmental Cell, 57*(3), 298-309.

12. Zhang, X., et al. (2021). The contribution of extrachromosomal DNA to genome plasticity in malaria parasites. Molecular Microbiology, 115(4), 503-507.

13. Quackenbush, R. L., et al. (1986). Extrachromosomal elements of Paramecium and their extrachromosomal elements. Basic Life Sciences, 40, 265-278.

14. Swart, E. C., et al. (2013). The Oxytricha trifallax macronuclear genome: a complex eukaryotic genome with 16,000 tiny chromosomes. PLoS Biology, 11(1), e1001473.

15. Hakim, A., Mor, Y., Toker, I. A., et al. (2018). WorMachine: machine learning-based phenotypic analysis tool for worms. *BMC Biology, 16*, 8.

16. Teichman, G., Cohen, D., Ganon, O., et al. (2023). RNAlysis: analyze your RNA sequencing data without writing a single line of code. *BMC Biology, 21*, 74.